

Final Exam Section Review— PSTAT 5LS

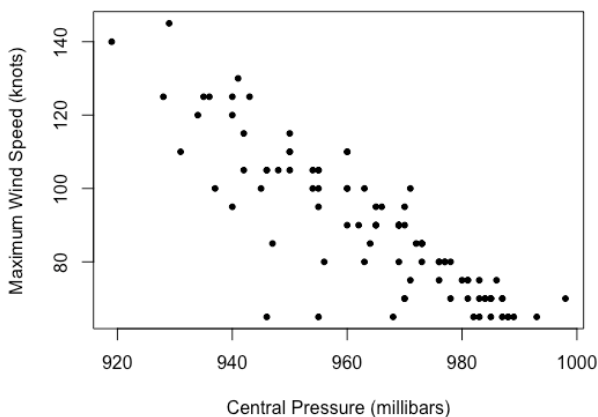
Spring 2026

A Note on Formatting: Unlike with Exams 1 and 2, the Final Exam will have questions that combine multiple choice and free response parts. The review problems below have been modified to reflect this format.

Use the following information for Questions 1 to 8:

Meteorologists study the relationship between a hurricane's **central pressure** (measured in millibars, mb) and its **maximum sustained wind speed** (measured in knots, kt). The following is a scatterplot and regression output for the linear model used to predict wind speed from central pressure.

Scatterplot of Wind Speed and Central Pressure



Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1013.11823	54.47291	18.60	< 2e-16 ***
central pressure	-0.95732	0.05645	-16.96	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.745 on 97 degrees of freedom

Multiple R-squared: 0.7478, Adjusted R-squared: 0.7452

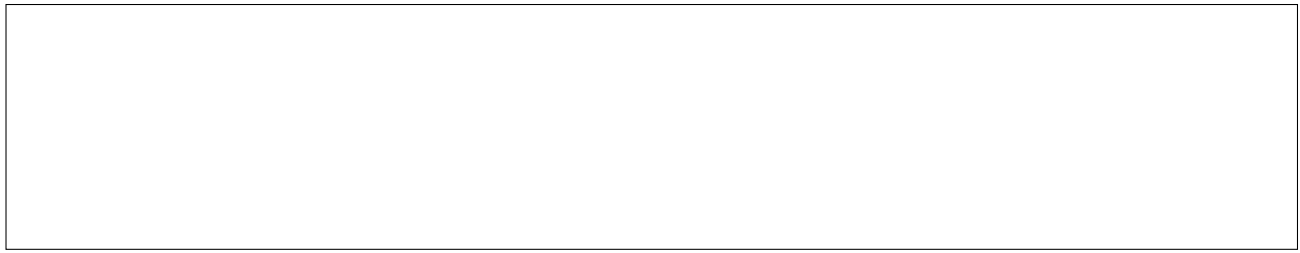
F-statistic: 2287.6 on 1 and 97 DF, p-value: < 2e-16

- Based on the regression results provided, which of the following is the best interpretation of the slope of the least-squares regression line?
 - ☐ For every 1 millibar increase in central pressure, the predicted maximum wind speed increases by about 1013.1 knots, on average.
 - ☐ For every 1 millibar increase in central pressure, the predicted maximum wind speed decreases by about 0.957 knots, on average.
 - ☐ For every 1 knot increase in maximum wind speed, the predicted central pressure decreases by about 0.957 millibars, on average.
 - ☐ For every 1 knot increase in maximum wind speed, the predicted central pressure increases by about 1013.1 knots, on average.
 - ☐ Approximately 95.7% of the variation in maximum wind speed can be explained by the change in central pressure.
- Calculate the value of the correlation coefficient r . Round to four decimal places.

3. Suppose the value of r was found to be $r = -0.9123$ (which is not necessarily the correct answer to the previous question!) Which of the following is the most appropriate description of this value in the context of the study?
- ☐ The value of -0.9123 indicates that as central pressure increases, the maximum wind speed also tends to increase.
 - ☐ If units for wind speed were converted from knots to kilometers per hour, the value of r would be multiplied by the appropriate conversion factor.
 - ☐ The value of -0.9123 suggests a strong, negative, linear relationship between central pressure and maximum wind speed.
 - ☐ A correlation of -0.9123 indicates that a decrease in central pressure is the direct cause of an increase in maximum wind speed.
4. Suppose a new hurricane is added to the data set with a central pressure of 990 mb and a maximum sustained wind speed of 120 knots (an outlier that does not follow the current trend). If this point were included in the regression analysis, how would it most likely affect the model?
- ☐ The slope would increase (become less negative/flatter) because the regression line is “pulled” toward this outlier.
 - ☐ The slope would decrease (become more negative/steeper) to compensate for the unusually high wind speed.
 - ☐ The correlation coefficient (r) would become closer to -1 because adding more data always strengthens a linear relationship.
 - ☐ The y -intercept would remain exactly the same because outliers only affect the slope of a line.
5. Write the equation of the regression line for predicting the average temperature from elevation. Be sure to define your variables clearly in the context of this problem.

6. Would it be proper to use the regression line to predict the maximum wind speed of a hurricane with a central pressure of 500 millibars? If so, predict the maximum wind speed of such a hurricane. If not, explain why this prediction problem is improper.

7. Would it be proper to use the regression line to predict the maximum wind speed of a hurricane with a central pressure of 940 millibars? If so, predict the maximum wind speed of such a hurricane. If not, explain why this prediction problem is improper.

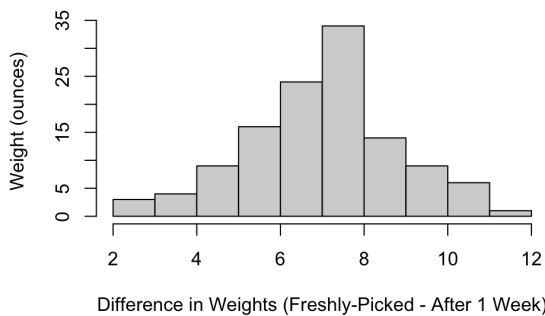


Use the following information for Questions 8 to 13:

Pear-ed Data

Farmer John grows Bartlett pears in his orchard. One day, his friend, Farmer Janice, told him that the average weight of a Bartlett pear decreases by 6.7 ounces after it's been sitting on a table for a week. Farmer John thinks that pears actually lose (on average) *more* weight than 6.7 ounces after sitting on a table for a week. To that end, Farmer John collects a random sample of 120 Bartlett pears from his tree and weighs them right after picking them. He then lets the pears sit on his table for a week, and reweighs the pears. Shown below is a summary of his data, and some potentially relevant R code.

Histogram of Differences in Pear Weights



```
> summary(pear_weights$weight_diff)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  2.381   5.827   7.133   6.999   8.008  11.064

> sd(pear_weights$weight_diff)
[1] 1.774222
```

8. Why is the parameter for this problem μ_d , the difference (freshly picked - after 1 week, in ounces) in pear weights and not $\mu_1 - \mu_2$?
- ☐ Because the sample of pear weights is random, the data is paired.
 - ☐ Because the weight measurements (freshly-picked and 1 week later) were taken on the same pear, the data is paired.
 - ☐ The specific choice of alternative hypothesis used in this problem results in paired data.
 - ☐ Because the distribution of differences in pear weights is roughly normal, the data is paired.

9. Check the conditions necessary for inference.

10. What are the null and alternative hypotheses?

- ☐ $H_0 : \mu_d = 6.7$ vs $H_A : \mu_d \leq 6.7$
- ☐ $H_0 : \mu_d = 6.7$ vs $H_A : \mu_d < 6.7$
- ☐ $H_0 : \mu_d = 6.7$ vs $H_A : \mu_d \neq 6.7$
- ☐ $H_0 : \mu_d = 6.7$ vs $H_A : \mu_d > 6.7$
- ☐ $H_0 : \mu_d = 6.7$ vs $H_A : \mu_d \geq 6.7$

11. Calculate the value of the test statistic. Show all your work, and round your final answer to three decimal places.

12. Which of the following correctly calculates the p -value? (In all choices below, “test statistic” is a placeholder for the correct value of the test statistic.)

- ☐ `pt(test statistic, df = 120, lower.tail = TRUE)`
- ☐ `pt(test statistic, df = 120, lower.tail = FALSE)`
- ☐ `pt(test statistic, df = 119, lower.tail = TRUE)`
- ☐ `pt(test statistic, df = 119, lower.tail = FALSE)`
- ☐ `2 * pnorm(test statistic, lower.tail = FALSE)`

13. The p -value is around 0.0337. Formulate a conclusion to Farmer John's hypotheses, using an $\alpha = 0.05$ significance level. Your conclusion should include:

- Your decision to reject or fail to reject the null hypothesis (and why)
- Whether or not the results are statistically significant
- A clear interpretation of the result in the context of the study

Use the following information for Questions 14 to 16:

Researchers want to study if daily meditation improves sleep quality among college students. They post announcements in several university dorms inviting students to participate. A total of 250 students sign up for the study and are then randomly assigned to either a meditation group (who practice 10 minutes of guided meditation daily) or a control group (who do not meditate daily). After four weeks, researchers compare the average sleep quality scores and find that the meditation group's average sleep quality scores improved compared to the control group.

14. Can the results of this study be generalized to the population of all students at the university?
- ☐ Yes, because the study uses random assignment to control and treatment groups.
 - ☐ Yes, because the sample includes a diverse group of students from multiple dorms.
 - ☐ No, because the sample size is too small to represent the entire student population.
 - ☐ No, because the study did not include students from off-campus housing.
 - ☐ No, because students who volunteer for the study might not be representative of the entire student population.
15. Which statement best describes the conclusions that can be drawn from this study?
- ☐ The study can establish a cause-and-effect relationship between daily meditation and improved sleep quality simply because the average sleep quality differs between the two groups.
 - ☐ The study can establish a cause-and-effect relationship between daily meditation and improved sleep quality because students were randomly assigned to treatment groups.
 - ☐ The study cannot establish a cause-and-effect relationship because participants were not randomly selected from the entire student population.
 - ☐ The study shows a correlation between meditation and sleep quality but cannot determine which causes which.
 - ☐ The study cannot establish causality because the study did not control for participants' prior sleep habits.
16. The researchers constructed a 92% confidence interval for $\mu_1 - \mu_2$, the difference (meditation – control) in average (mean) sleep scores, and found it to be (9.021 , 11.203). What is the value of $\bar{x}_1 - \bar{x}_2$, the sample mean sleep score in the meditation group minus the sample mean sleep score in the control group? Show your work, and round to three decimal places.